

GEOMETRIC PHOTONIC RAM (G-PRAM): A Hybrid Photonic–Plasmonic Metasurface Architecture for Frequency-Shift In-Memory Computing

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Abstract

The von Neumann bottleneck cripples large-scale AI accelerators, with data movement devouring up to 80% of system energy. Geometric Photonic RAM (G-PRAM) fuses non-volatile memory and nonlinear logic in each metasurface unit cell: high-Q silicon nitride (SiN) resonators for readout, plasmonic nano-antennas for sub-diffraction writes, and thin-film lithium niobate (TFLN) for $\chi^{(2)}$ frequency-shift outputs. Analytical models forecast switching energies of 10–50 fJ (Gen-1) to sub-10 fJ (Gen-2), endurance $> 10^7$ cycles, and dense WDM parallelism using near-term nanofabrication [1, 2, 3].

1 Introduction

Photonic platforms smash resistive losses and enable WDM/ultrafast nonlinearities, but photonic memories still wrestle with energy, endurance, and density trade-offs [4]. G-PRAM integrates low-loss Sb_2Se_3 PCM for bit encoding via phase/geometry; plasmonic hotspots drive writes; TFLN SHG handles decoupled, frequency-encoded readout [1, 10, 11].

2 Hybrid Unit Cell Architecture

The hybrid unit cell consists of a tripartite stack:

- SiN resonator: $Q > 10^6$, low-loss spectral filtering [12].
- Au/Ag plasmonic dimer: ~ 10 nm hotspot for write confinement, minimal readout Q impact [17].
- Heterogeneous TFLN layer: high $\chi^{(2)}$ for SHG [3, 22]. Bondable diamond substrate sinks heat [23, 24].

3 Switching Energy Model

Sb_2Se_3 phase transition driven by optical-thermal heating of the active volume:

$$E = V \cdot \rho \cdot c \cdot \Delta T + V \cdot \rho \cdot L \cdot \eta^{-1} \quad (1)$$

where:

- V : heated PCM volume ($< 10^{-24} \text{ m}^3$ via plasmonic hotspot),

- ρ : Sb_2Se_3 density,
- c : specific heat,
- ΔT : crystallisation threshold from ambient [25],
- L : latent heat of phase change,
- η : coupling/absorption efficiency (0.3–0.6 typical in hybrids) [26].

Sb_2Se_3 delivers $\Delta n \approx 0.77$ with ultra-low loss (< 0.01 dB/ μm both phases at 1550 nm) [1, 2].

3.1 Realistic Estimates

Full-volume thermal budget ~ 900 fJ baseline; plasmonic confinement + partial switching shaves $10\text{--}50\times$ [30]. Patterned nano pillars and graphene thermal barriers push endurance $> 10^7$ cycles [34].

3.2 Generational Targets

Generation	Energy	Key Enablers
Gen-1	10–50 fJ	Plasmonic– Sb_2Se_3 hybrids, partial volume [1, 2]
Gen-2	< 10 fJ	Graphene assist + optimised hotspots [34]

Table 1: Generational performance targets.

4 Readout Mechanisms

4.1 Gen-1 (Intensity-Based)

Probe at fundamental wavelength; high transmission (“1”) amorphous/on-resonance, low (“0”) crystalline/detuned [10].

4.2 Gen-2 (SHG Frequency-Shift)

Non-linear output via TFLN:

$$P_{\text{SHG}} \propto \gamma |E_{\text{pump}}|^4 \quad (2)$$

where γ scales with cavity enhancement. Recent domain-engineered TFLN waveguides hit 2590 %/W SHG efficiency with high pump depletion [3, 22, 38]; μW probes deliver logic without excessive heat.

5 Thermal Dissipation

Steady-state hotspot rise under average power:

$$\Delta T \approx \frac{P_{\text{avg}}}{2\pi\kappa r} \quad (3)$$

with $\kappa \approx 2000$ W/m $\cdot$ K (bondable diamond [23, 24]), $r \approx 50$ nm effective radius. At 50 fJ/bit and 10 GHz, $P_{\text{avg}} \approx 0.5$ mW $\rightarrow \Delta T \approx 5$ K, minimising crosstalk [42].

6 In-Memory Photonic Computation

Spatially modulated wavefronts interact with PCM-weighted metasurface for linear transforms; SHG provides frequency-multiplexed non-linear outputs. Inverse-designed Sb_2Se_3 elements enable reconfigurable diffractive networks [46].

7 Fabrication Roadmap (2026–2030)

Feature	Gen-1	Gen-2	Refs
Energy	10–50 fJ	< 10 fJ	[1, 34]
Readout	Intensity	SHG frequency logic	[3, 22]
Density	4.6×10^8 bits/cm ²	1.8×10^9 bits/cm ²	—
WDM Channels	16–32	128–256	[12]
Endurance	> 10^7 cycles	> 10^8 (patterned + graphene)	[34]
Architecture	Tiled modules	Monolithic ~ 38 cm ²	[23, 24]

Table 2: Fabrication and performance roadmap.

8 Conclusion

G-PRAM is a manufacturable blueprint for unitary photonic AI: Sb_2Se_3 low-loss switching, TFLN non-linear readout, diamond thermal mastery—sub-50 fJ and high parallelism within reach of current foundry flows [1–18].

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